

Question number	Answer	Additional guidance	Mark
4(a)	A 1 is the correct answer B not 2 as only the collecting duct is affected by ADH C not 3 as only the collecting duct is affected by ADH D not 4 as only the collecting duct is affected by ADH		(1)

Question number	Answer	Additional guidance	Mark
4(b)(i)	ultrafiltration	Accept glomerular ultrafiltration	(1)

Question number	Answer	Additional guidance	Mark
4(b)(ii)	microvilli		(1)

Question number	Answer	Additional guidance	Mark
4(c)(i)	A calculation that has the following stages: - Correctly calculating the radius (1) - Correctly calculating the cross-sectional area (1)	2.355, accept 2.36 17.43, accept 17.5 PE to determine suitable range at pre-standardisation.	(2)

Question number	Answer	Additional guidance	Mark
4(c)(ii)	Correct calculation of the difference, must include the correct units	1.97, difference in mm ² 0.0197, difference in cm ² , accept 0.02. PE to determine suitable range at pre-standardisation	(1)

Question number	Answer	Additional guidance	Mark
4(d)	An answer that includes three of the following points: - proximal tubule has role in selective reabsorption (1) - named relative concentration change by reabsorption (1) - comment on protein channels for reabsorption (1) - named relative concentration change by secretion (1)	e.g. 100% of glucose e.g. 70x increase for urea	(3)

Question number	Answer	Additional guidance	Mark
4(e))	A two-step calculation: - Calculation of urine in one hour (1) - Correct total for 24 hours to two s.f. (1)	(0.83 x 60 = 49.8) 1.2dm ³	(2)

